CS224 Object Oriented Programming and Design Methodologies Quiz 1

Quiz Guidelines

You will attempt this quiz on your lab machines. Your submission should be a single cpp file named `firstname lastname quiz1.cpp`. The file should contain all of the code and implementation of the quiz problems. You may use the following documentation to look up syntax:

https://en.cppreference.com/w/.

Accessing any site other than HULMS and the provided documentation is prohibited. You are not allowed to communicate with your peers or use any external resources. You cannot refer to your notes, books or lecture slides during the quiz. Use of generative AI is strictly prohibited. Not abiding by any of the rules will result in an immediate zero in the quiz grade and a possible academic conduct case.

Problem – Enchanted Scroll

During her travels, Frieren, the Elven Mage, discovers an enchanted scroll hidden inside a mimic. The scroll is sealed with n binary values. To dispel the magic, Frieren must:

1. Identify the binary number with the largest amount of mana, that is, the binary number with the largest decimal value.
2. Identify the binary number with the smallest amount of mana, that is, the binary number with the smallest decimal value.
3. Perform the Zoltraak spell on these two binary numbers that is, apply a bitwise AND operation on these two binary numbers.

Since Frieren isn't very good at mathematics, you must help her by writing the following functions:

ConvertBinary

- Takes an integer array of size 8 (containing only 0s and 1s).
- The array represents a binary number in reverse order (the least significant bit is at index 0).
- Returns the decimal value of this binary number.

Example: For the binary number `10010110`, the decimal value is:

$$2^0 + 2^3 + 2^5 + 2^6 = 1 + 8 + 32 + 64 = 105$$

PrintValues

- Takes the integer 2D array and the integer parameter n (number of binary values).
- For each binary number, prints both the binary value and its decimal equivalent (using `ConvertBinary`).

GetMaxValue

- Takes the integer 2D array and the integer parameter n.
- Finds the binary number with the largest decimal value.
- If multiple binary numbers share the maximum value, select the first occurrence.
- Returns the index of this binary number.

GetMinValue

- Takes the 2D array and the integer parameter n.
- Finds the binary number with the smallest decimal value.
- If multiple binary numbers share the minimum value, select the first occurrence.
- Returns the index of this binary number.

Zoltraak

- Takes two integer arrays of size 8 (each containing only 0s or 1s).
- Applies the bitwise AND operation between them.
- Returns the resulting binary number.

Main

- Prompts the user to enter the number of binary strings n.
- For each binary string, prompts the user to enter its 8 bits (0 or 1), starting from the least significant bit (which will be stored at index 0).
- Stores all values in an integer 2D array. As shown:

$$[[1, 0, 0, 1, 0, 1, 0], [1, 1, 0, 0, 1, 1, 0, 1], [0, 1, 0, 1, 0, 1, 0, 1]]$$

- Calls `PrintValues` to display each binary number and its decimal value.
- Calls `GetMaxValue` and `GetMinValue` to display the largest and smallest binary strings.
- Calls `Zoltraak` on the largest and smallest binary strings and displays the result.

Test Case 1

Inputs:

```
3
1 0 1 1 0 1 0 0
1 1 0 0 1 1 0 1
0 1 0 1 0 1 0 1
```

Outputs:

```
Binary string 0 = 10110100 = 45
Binary string 1 = 11001101 = 179
Binary string 2 = 01010101 = 170
Largest value: index 1 -> 11001101
Smallest value: index 0 -> 10110100
AND result: 10000100
```

Test Case 2

Inputs:

```
4
1 0 0 1 0 0 1 0
0 1 1 0 1 1 1 1
0 0 0 1 1 1 0 0
1 1 0 0 1 1 0 0
```

Outputs:

```
Binary string 0 = 10010010 = 73
Binary string 1 = 01101111 = 246
Binary string 2 = 00011100 = 56
Binary string 3 = 11001100 = 51
Largest value: index 1 -> 01101111
Smallest value: index 3 -> 11001100
AND result: 01001100
```